

AI Usage DISCLOSURE FORM

1. Team Details

Team Name: SRMIST_VectorFlow

Project / Product Name: Streaming Live RAG Engine (Theme 04: Real-Time Incremental Retrieval, Multi-Intent Decomposition, and State-Preserving Answer Refinement)

Organization / Institution (if any): SRM Institute of Science and Technology (SRMIST), Kattankulathur

Submission Date: 30 September, 2026

2. AI Usage Declaration

Did your team use any Artificial Intelligence (AI) in developing this project? Yes / No

- Yes
- **Declaration:**
- AI tools were utilized during development for code scaffolding, boilerplate generation, UI layout prototyping, and test fixture synthesis. Additionally, AI/LLM models form an integral, purposeful component of the runtime system architecture (specifically for sub-query decomposition, query understanding, and state-preserving grounded answer synthesis under strict corpus isolation).

3. Purpose of AI Usage (Brief Details)

Idea generation / brainstorming - Structuring failure-mode edge cases, drafting architectural trade-offs, and defining acceptance benchmarks for evaluation gates G1-G6

Code generation or assistance - Generating initial Python boilerplate for FastAPI endpoints, Pydantic v2 data models, and rank-fusion mathematical formulas.

UI / UX design - Prototyping CSS glassmorphism styles and layout scaffolding for the 3-panel real-time streaming dashboard.

Content creation - Drafting system documentation, reproducible README instructions, and the System Architecture Brief.

Data analysis - Aggregating runtime telemetry data, computing latency percentiles (P95), and formatting ablation comparison tables.

Testing / debugging - Generating synthetic timestamped transcript replay fixtures and designing scenario tests for late-arriving constraints and query suppression.

Other (Runtime Inference): Serving as the core runtime cognitive engine for intent decomposition and citation-grounded answer generation.

4. Feature Origin Classification (Please add/delete based on the number of the feature)

1. Feature Name - Fast-Path Retrieval Controller (C1)

2. Self-Generated/AI-Generated/Both - Both

3. Description (include AI tools/platform Used, prompt used , Output Summary, Modification)

AI Tools / Platform Used: Google Antigravity / Claude Sonnet / Gemini Pro

Implement a sub-15ms deterministic controller using regex for presentation turns ('turn into bullets') and syntactic stability heuristics (entity + verb presence) to emit WAIT, RETRIEVE, or SUPPRESS.

Fast decision algorithms, regex pattern recognizers, and decision event schemas

1. Feature Name - Multi-Intent Query Decomposer & Parallel Routing (C2)

2. Self-Generated/AI-Generated/Both - Both

3. Description (include AI tools/platform Used, prompt used , Output Summary, Modification)

AI Tools / Platform Used: Google Antigravity & LLM APIs (GPT-4o-mini / Gemini Flash)

Prompt Used: "Generate a structured JSON-schema extractor that isolates distinct orthogonal sub-queries from compound spoken transcripts without over-fragmenting overlapping intents.

& many more

5. Ethical & Compliance Confirmation

AI usage complies with guidelines and policies. **Yes**

No proprietary or copyrighted data misused. **I Agree**

6. Declaration & Sign-Off

Name of Team Representative: Parth Tewari

Role: Team Lead / Full-Stack & Systems Engineer

Signature: Parth Tewari

Date: 30 September 2026